Stockd.

Software Development Plan (SDP)

CMSI 4071 — Senior Project I | Fall 2026

Mason Lui • Leila Nawas • Paige Inoue

4.0 Software Development Plan

4.1 Plan Introduction

Stockd. is a smart retrofit refrigerator system that helps people keep track of the food they currently have. Our goal is to use camera and food recognition softwares to notice when food is added or removed from a refrigerator and update virtual inventories accordingly. Users will be able to check their inventory from their phone, see expiration reminders, and get recipe suggestions based on available ingredients.

We plan to build Stockd. in multiple stages. First, we want to make a phone-camera version so we can test the app and food recognition without buying hardware. We will have this preliminary version connected with our mobile app or website that we developed and check all UI and backend software. Once those both work, we will connect a Raspberry Pi camera and magnetic door sensor to a small demonstration refrigerator. Pantry tracking is an optional feature and is not part of our required first version.

We want to use an iterative development approach similar to Agile. We will work on a small group of features, test them, and make improvements before moving on to the next steps. This is more practical for our project than waiting until the very end to test everything since there are both hardware and software components that must work together.

4.2 Project Resources

4.2.1 Hardware Resources

Our laptops and phones will be enough for the first prototype. We plan to purchase the refrigerator hardware later in the semester after the phone-based system works.

Hardware	How we will use it
Team laptops	Write code, test the app, and update documents
Smartphone	Take food photos for the early prototype
Raspberry Pi 5 (4 GB)	Run the camera program and communicate with the backend
Camera Module 3 Wide	Capture images of food in the demonstration fridge
Magnetic door sensor	Tell the Pi when the refrigerator door opens and closes
Small demonstration refrigerator	Use for our physical Stockd. demo
Power supply, microSD, camera cable, mounts	Connect and power the equipment

4.2.2 Software Resources

These are the tools we currently expect to use. Some may change if we discover a better or simpler option while building the project.

Software	Purpose
React, TypeScript, Vite	Build the web app for phones and computers
Python / FastAPI	Handle image uploads and communicate with the app
Supabase (PostgreSQL)	Store users, food items, expiration dates, and inventory history
Vision API / recognition model	Identify foods in uploaded images
OpenCV / Picamera2	Handle motion checks and camera images on Raspberry Pi
GitHub	Store code, track changes, and submit documents

Cursor / VS Code	Develop and debug our code

4.3 Project Organization

Between our three member team, Mason Lui, Leila Nawas, and Paige Inoue, we plan to split tasks based on what each person is comfortable working on and will help each other as needed or when stuck. We have not yet determined who will work on which specific areas of the project yet, but we plan to collaborate on each part in order for ideas to be shared and feedback to be heard between each other.

Work area	Main tasks
Frontend	Dashboard, inventory pages, photo upload, recipe page
Backend / AI	Food recognition, image comparisons, database updates
Hardware	Raspberry Pi camera, magnetic sensor, refrigerator setup
Testing / documentation	Check features, update GitHub, write reports, prepare demos

For our group, communicating regularly, pushing our changes to GitHub, and reviewing each other's work before combining major features are all things we agree to prioritize and practice during the development of our project. Our issues or shared task list will help us keep track of what is finished and what still needs work.

Development Process

We plan to use short development cycles throughout the development process. At the start of each cycle we will choose a few tasks, build them, test them, and share the results with each other. A feature will count as done when it is fully integrated and working in the app, its main error cases have been tested, and the code is saved in our GitHub repository.

Testing and Quality

Testing features as we build them instead of waiting until the end is crucial for our team's project plan. Some features include making sure users can edit inventory items, scan images, correct AI mistakes, see expiration alerts, and get recipe recommendations which are important for the overall function of our product. We will also test camera capture and door detection once we have the hardware fully integrated and set up. Because food recognition can make mistakes, users will be able to confirm uncertain changes in the mobile app and/or website.

Main Risks

Risk	Our plan
AI does not identify food correctly	Test a limited set of foods first and allow users to correct results
Hardware delivery or setup takes longer	Finish and demonstrate the phone-camera version first
Camera cannot see all items clearly	Try different mounting positions and lighting
Team falls behind schedule	Prioritize inventory, recognition, and recipes; postpone optional pantry tracking

4.4 Project Schedules

Our schedule is based on the course weeks rather than exact dates. We expect to adjust it as we test features. The course deadlines are fixed, but the development tasks below are estimates.

4.4.1 PERT / GANTT Chart

An X indicates the weeks when we expect to work on a task.

Task	7	8	9	10	11	12	13	14	15	16

App and database setup	X	X	X							
Phone photo scans		X	X	X						
Inventory / expiration			X	X	X	X				
Recipe recommendations				X	X	X				
Architecture / design docs			X	X	X	X				
Raspberry Pi / door sensor					X	X	X			
Testing and fixes						X	X	X	X	
Presentation / final hand-in							X	X	X	X

4.4.2 Task / Resource Table

Task	Needed resources	Target
Finish SDP	All team members, documents	Week 7
Basic app and inventory	Laptops, React, Supabase	Weeks 8–10
Phone image recognition	Phone, backend, vision service	Weeks 9–10
Expiration and recipes	Database, frontend, recipe logic	Weeks 10–12
Physical camera setup	Pi, camera, door sensor, mini fridge	Weeks 11–13
Testing and demonstration	All team members, test food items	Weeks 12–15
Final files and presentation	GitHub, README, presentation	Weeks 15–16

4.4.3 Class Schedule

Due week	Course deliverable
Weeks 2–3	D#01 Project Proposal
Week 5	D#02 Requirements Specification and Initial Schedule
Week 7	D#03 Complete Software Development Plan
Week 10	D#04 Architecture Design Document
Week 12	D#05 Detailed Design Document
Weeks 12–13	D#06 Alpha/Beta/Critical Design Review and Demonstration
Week 13	D#07 README User Manual Updates
Week 15	D#08 Final Presentation
Week 16	D#09 Final Report, Code, and Documentation
Weeks 7–15	D#10 Oral Status Reports
Every other week from Week 6	D#11 Written Status Reports

We will compare these plans with the project schedule already in our GitHub repository and update the timeline when major tasks change.